

Software Manual: Habitat imaging-derived “solid component” for predicting high-risk lung adenocarcinomas in part-solid nodules on low-dose CT

1. Software Overview

1.1 Software Introduction

This software implements a fully automated habitat imaging pipeline for quantifying solid components within part-solid nodules on low-dose computed tomography (LDCT) and predicting high-risk lung adenocarcinoma (LUAD). Using fixed thresholds (density: -447 HU; entropy: 4.201), density and entropy maps are clustered to generate four distinct habitats. The solid component is defined as the **high-density and low-entropy** region (Label 3), and its volume together with its volume ratio are used as key predictors in a logistic regression model.

1.2 Key Features

- **Single Case Processing:** Analyze CT images and segmentation masks for individual patients, including manual sex selection (man/woman).
- **Batch Processing:** Automatically process multiple cases with automatic file pairing; batch sex mapping via a CSV file (ID, Sex).
- **Habitat Imaging Pipeline:** Six-step workflow (resample $\rightarrow$ entropy maps $\rightarrow$ threshold masks $\rightarrow$ fusion masks $\rightarrow$ volume statistics $\rightarrow$ logistic regression prediction) based on the thresholds derived from the training set.
- **Interpretable Visualization:** Outputs a 4-label image mapping distinct habitats with color coding.
- **Result Visualization:** Intuitive display of label volumes and prediction results with color coding (red = high-risk, green = low-risk).
- **Result Export:** Export prediction results, habitat volumes, probability, and sex to CSV format.

1.3 Software Interface

The software features a tabbed graphical user interface with four main sections:

Tab	Description
 Input & Processing	File selection, mode choice (single/batch), parameter input, sex input (single case or CSV for batch), processing controls, and real-time log display
 Single Case Results	Detailed results of the last processed single case, including volumes of four habitat labels and final prediction
 Batch Results	Summary table of all processed cases in batch mode, with sex, label volumes, probability, prediction, and processing status
 About	Software version, description, processing steps, thresholds, logistic regression formula, and prediction threshold

2. File Requirements and Format

2.1 Supported File Formats

- **CT Images:** .nii.gz or .nii (NIfTI format)
- **Segmentation Masks:** .nii.gz or .nii (NIfTI format)

2.2 File Naming Conventions

The software automatically matches CT and mask files. The required naming pattern is:

- **CT file:** casename.nii.gz
- **Mask file:** casename_mask.nii.gz

Example:

CT File	Mask File
patient001.nii.gz	patient001_mask.nii.gz
subject_123.nii	subject_123_mask.nii

If files follow this convention, they will be correctly paired during batch scanning.

2.3 Batch Sex Mapping File (for Batch Processing)

When running batch processing, a CSV file containing sex information for each case must be provided.

The CSV must contain exactly two columns:

- **First column:** Case (the base filename without _mask.nii.gz extension, e.g., patient001).
- **Second column:** Sex (0 for man, 1 for woman).

Example (sex.csv):

Case	Sex
patient001	0
patient002	1

3. Software Operation Guide

3.1 Launching the Software

Double-click the executable Habitat_LUAD.exe. The main window will appear.

3.2 Processing Modes

3.2.1 Single Case Processing

1. Select "**Single Case**" radio button.
2. Click "**Browse CT**" to select the CT file.
3. Click "**Browse Mask**" to select the corresponding mask file.
4. Choose **Sex** from the dropdown: man (0) or woman (1).
5. Configure output directory (default: C:\Users\YourName\HabitatLUAD_Output).
6. The density threshold (–447 HU) and entropy threshold (4.201) are fixed; they are not editable.
7. Click "**Start Processing**". The log will show progress; upon completion, the **Single Case Results** tab opens automatically.

3.2.2 Batch Processing

1. Select "**Batch Processing**" radio button.
2. Click "**Browse Directory**" to select the folder containing all CT and mask files.
3. Click "**Browse CSV**" to select the sex mapping CSV file, then click "**Load Sex Map**".
4. Click "**Scan for File Pairs**" – the software will list all detected valid pairs (green check marks).
5. Review the detected file pairs in the list.
6. Configure output directory as needed (thresholds are fixed).
7. Click "**Start Processing**". The progress bar updates for each case; results appear in the **Batch Results** tab in real time.

3.3 Common Settings

- **Output Directory:** Location where results for each case will be saved (subfolders are created automatically).
- **Density Threshold (HU):** Fixed to –447 HU (derived from the training set).
- **Entropy Threshold:** Fixed to 4.201 (derived from the training set).
- **Prediction Model:** Logistic regression using sex, Label 3 volume (converted to cm³), and

Label 3 volume ratio. The probability threshold for high-risk classification is 0.171.

4. Processing Pipeline

The software executes the following six steps for each case:

4.1 Step 1: Image Resampling

- **Description:** Both CT and mask images are resampled to isotropic 1 mm³ voxel spacing to standardize spatial resolution.
- **Output:** Saved in 1_resampled/ directory.

4.2 Step 2: Local Entropy Calculation

- **Description:** A 3D local entropy map is computed after applying lung window normalization (width = 1500, level = -500) to the resampled CT image. A 3×3×3 moving window is used.
- **Output:** Entropy map saved in 2_entropy_maps/ directory.

4.3 Step 3: Generation of Four Threshold Masks

- **Description:** Within the original mask region, four binary masks are created by applying the density and entropy thresholds:
 - **High Density:** CT values ≥ -447 HU
 - **Low Density:** CT values < -447 HU
 - **High Entropy:** Entropy values ≥ 4.201
 - **Low Entropy:** Entropy values < 4.201
- **Output:** Four masks saved in 3_threshold_masks/ directory.

4.4 Step 4: Fusion into a 4-Label Image

- **Description:** The four binary masks are combined pixel-wise to create a single 4-label image according to the following rule:
- Label 1: Low density & Low entropy → Green
- Label 2: Low density & High entropy → Blue
- Label 3: High density & Low entropy (solid component) → Red
- Label 4: High density & High entropy → Yellow
- **Output:** Fused label image saved in 4_fusion_masks/ directory.

4.5 Step 5: Volume Statistics

- **Description:** For each of the four labels, the number of voxels, volume (mm³), and volume ratio (relative to the total mask volume) are calculated.
- **Note:** Because of prior resampling to 1 mm spacing, each voxel corresponds to 1 mm³.
- **Output:** Statistics are displayed in the log and used for the next step.

4.6 Step 6: Visualization Generation

- **Description:** After prediction, a composite figure is automatically generated to visually summarize the analysis. This figure displays:
 - CT map (lung window)
 - Entropy map
 - Habitat labels (with color legend)
 - Solid component (Label 3) with annotation showing its volume (mm³), ratio (%), probability, and final prediction.
- **Output:** Saved as 05_composite.png in the 5_visualizations/directory. Single images (01_ct_map.png, 02_entropy_map.png, 03_habitat_labels.png, 04_solid_component.png) are also saved in the same folder.

4.6 Step 7: High-Risk LUAD Prediction

- **Description:** Using the volume of Label 3 (high-density & low-entropy) in cm^3 and its volume ratio, together with the patient's sex, a logistic regression model calculates the probability of high-risk LUAD.
- **Formula:** $\text{logit} = -2.808 - 0.697 \times \text{sex} + 1.702 \times \text{Label3_volume (cm}^3) + 0.018 \times \text{Label3_ratio}$
 $\text{probability} = 1 / (1 + \exp(-\text{logit}))$, where sex = 0 for man, 1 for woman.
- **Decision Rule:**
 - High-risk LUAD (Class 1): probability > 0.171
 - Low-risk LUAD (Class 0): probability $\leq$ 0.171
- **Output:** Prediction result, probability, and volumes saved as a CSV file in 6_prediction_results/ directory.

5. Output Structure

For each processed case, a subfolder named after the case is created in the output directory, containing the following files:

HabitatLUAD_Output/

```
└── CaseName/
    ├── 1_resampled/
    │   ├── CaseName_resample.nii.gz          (Resampled CT, 1mm isotropic)
    │   └── CaseName_resample_mask.nii.gz      (Resampled mask, aligned)
    ├── 2_entropy_maps/
    │   └── CaseName_resample_entropy.nii.gz   (3D entropy map after lung window)
    ├── 3_threshold_masks/
    │   ├── CaseName_high_density_mask.nii.gz
    │   ├── CaseName_low_density_mask.nii.gz
    │   ├── CaseName_high_entropy_mask.nii.gz
    │   └── CaseName_low_entropy_mask.nii.gz
    ├── 4_fusion_masks/
    │   └── CaseName_fused_labels.nii.gz       (4-label image: 1 - 4 with updated mapping)
    ├── 5_visualizations/
    │   ├── 01_ct_map.png
    │   ├── 02_entropy_map.png
    │   ├── 03_habitat_labels.png
    │   ├── 04_solid_component.png
    │   └── 05_composite.png
    └── 6_prediction_results/
        └── CaseName_prediction.csv           (Prediction result, probability, and label volumes)
```

5.1 Output File Descriptions

Directory	File Pattern	Description
1_resampled/	*_resample.nii.gz	Resampled CT image
	*_resample_mask.nii.gz	Resampled mask aligned with CT
2_entropy/	*_resample_entropy.nii.gz	Local entropy map
3_threshold_masks/	*_high_density_mask.nii.gz	Binary mask: $\text{CT} \geq -447 \text{ HU}$

	*_low_density_mask.nii.gz	Binary mask: CT < -447 HU
	*_high_entropy_mask.nii.gz	Binary mask: entropy ≥ 4.201
	*_low_entropy_mask.nii.gz	Binary mask: entropy < 4.201
4_fusion_masks/	*_fused_labels.nii.gz	4-label image
5_visualizations	*_ct_map.png	A representative axial slice of the resampled CT image (typically the central slice of the mask's bounding box).
	*_entropy_map.png	The corresponding slice of the 3D local entropy map.
	*_habitat_labels.png	Color-coded fused label image overlaid on CT slice (green/blue/red/yellow)
	*_solid_component.png	Highlighted Label 3 (high-density & low-entropy) region
	*_composite.png	Composition of the above images
6_prediction_results/	*_prediction.csv	CSV file containing case name, sex, volumes of four labels (mm ³), volume ratios (%), probability, and prediction (0/1)

6. Results Interpretation

6.1 Single Case Results Tab

After single-case processing, the **Single Case Results** tab displays a table with the following columns:

Column	Description	Format / Example
Case	Patient identifier (derived from filename)	patient001
Sex	man or woman	man
Label1 Volume	Volume of low-density & low-entropy region (mm ³)	2021
Label2 Volume	Volume of low-density & high-entropy region (mm ³)	333
Label3 Volume	Volume of high -density & low-entropy region (mm ³)	2503
Label4 Volume	Volume of high -density & high-entropy region (mm ³)	380
Label1 Volume ratio	Volume ratio of low-density & low-entropy region	38.6
Label2 Volume ratio	Volume ratio of low-density & high-entropy region	6.4
Label3 Volume ratio	Volume ratio of high -density & low-entropy region	47.8
Label4 Volume ratio	Volume ratio of high -density & high-entropy region	7.9
Probability	Predicted probability of high-risk LUAD (0–1)	0.910
Prediction	Final classification (based on probability > 0.171)	High-risk

- The Prediction cell is color-coded: **red** for high-risk, **green** for low-risk.

6.2 Batch Results Tab

Column	Description	Format / Example
Case	Patient identifier	patient002
Sex	man or woman	man
Label1-4 volumes	As above (mm ³)	125, 89, 214, 46
Probability	Predicted probability	0.823
Prediction	High-risk / Low-risk	High-risk
Status	Success / Error message	Success

- Prediction is color-coded similarly.
- Failed cases appear with a light red background and an error message in the Status column.

6.3 Prediction Model and Threshold

- **Formula:** $\text{logit} = -2.808 - 0.697 \times \text{sex} + 1.702 \times \text{Label3_volume (cm}^3\text{)} + 0.018 \times \text{Label3_ratio}$
 $\text{probability} = 1 / (1 + \exp(-\text{logit}))$, where sex = 0 for man, 1 for woman.
- **Decision Rule:**
 - High-risk LUAD (Class 1): probability > 0.171
 - Low-risk LUAD (Class 0): probability ≤ 0.171

The model uses Label 3 (high density & low entropy) as the solid component predictor, together with the patient's sex.

7. Disclaimer

7.1 Intended Use

This software at the current stage is primarily intended for **research purposes**. While it can provide valuable clinical references, it should be integrated with comprehensive clinical evaluation rather than used as the sole basis for clinical decision-making.

7.2 Limitations

- Requires accurate CT images and tumor masks; results are highly dependent on the quality of the input mask.
 - Performance may vary with different patient populations, CT scanner models, and acquisition protocols.
 - The fixed thresholds (-447 HU, 4.201) were derived from a specific study population; they may not be optimal for other cohorts without recalibration.
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8. Continuous Improvement

We are committed to the ongoing validation and improvement of this tool. Users are encouraged to:

- Provide feedback on clinical utility and performance.
- Participate in validation studies.
- Report any discrepancies with clinical outcomes.
- Share suggestions for enhancement.

For questions or feedback, please contact the software distributor or the original study authors.
